

Sec 3.9 – Linear Approximation and the Derivative

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

When we zoom in on the graph of a differentiable function, eventually it will look like a straight line, because the deviation from straightness can't be detected.

So given a differentiable function, we can use the tangent line approximation at a point $x=a$ as an approximation to the function near that point $x=a$.

Theorem

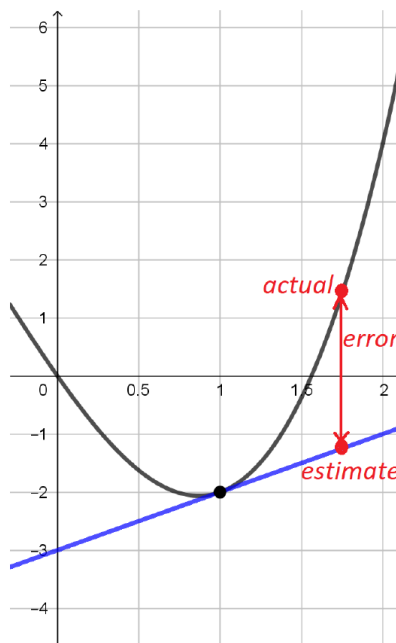
Suppose f is differentiable at a . Then, for values of x near a , the tangent line approximation to $f(x)$ is

$$f(x) \approx f(a) + f'(a)(x - a).$$

The expression $f(a) + f'(a)(x - a)$ is called the **local linearization of f near $x = a$** . We are thinking of a as fixed, so that $f(a)$ and $f'(a)$ are constant.

The **error**, $E(x)$, in the approximation is defined by

$$E(x) = f(x) - [f(a) + f'(a)(x - a)].$$



The **local linearization** of f near $x = a$ is just another name for the **tangent line** to f at $x = a$.

Compare:

$$f(x) \approx f(a) + f'(a)(x - a).$$

$y = m(x - a) + b.$

Arrows indicate the correspondence: $f(x)$ corresponds to y , $f(a)$ corresponds to b , $f'(a)$ corresponds to m , and $(x - a)$ corresponds to $(x - a)$.

- $f(x)$ is in place of y
- $f'(a)$ is in place of m (notice this is a *number*!)
- $f(a)$ is in place of b
- the order of the two terms on the right is switched

Find the tangent line to the function $f(x) = \sqrt[3]{x}$ at $x = 8$.

Use the tangent line to approximate $\sqrt[3]{8.1}$ (without a calculator)

How far off is our tangent line estimate from the actual value of $\sqrt[3]{8.1}$?

Solutions:

$$f'(x) = \frac{1}{3}x^{-2/3}, \text{ so } f'(8) = \frac{1}{3}(8)^{-2/3} = \frac{1}{12}$$

- Since $f(8) = 2$, the tangent line to $f(x)$ at $x = 8$ is $y = \frac{1}{12}(x - 8) + 2$.
- Using the tangent line, we estimate
 $f(8.1) \approx \frac{1}{12}(8.1 - 8) + 2 = \frac{1}{12}(.1) + 2 = 2\frac{1}{120} \approx 2.00833333$.
- The actual value is 2.0082988502, so the error is
 $2.0083333333 - 2.0082988502 = 0.075034483$.

Partner up

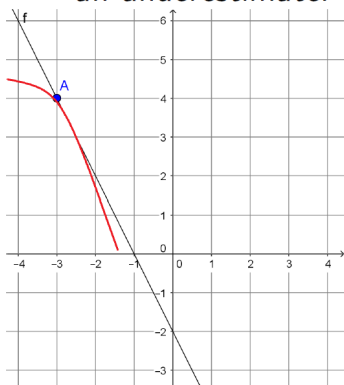
Example 2: Given a function $g(x)$ that is differentiable and $g(-3) = 4$, $g'(-3) = -2$, and $g''(-3) = -1$.

- 1 Find the local linearization of $g(x)$ at $x = -3$.
- 2 Use it to estimate $g(-3.1)$.
- 3 Draw a graph of $g(x)$ and determine if $g(-3.1)$ is an overestimate or an underestimate.

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Given a function $g(x)$ that is differentiable and $g(-3) = 4$, $g'(-3) = -2$, and $g''(-3) = -1$.

- 1 Find the local linearization of $g(x)$ at $x = -3$.
 $g(x) \approx -2(x + 3) + 4$
- 2 Use it to estimate $g(-3.1)$.
 $g(-3.1) \approx -2(-3.1 + 3) + 4 = 4.2$
- 3 Draw a graph of $g(x)$ and determine if $g(-3.1)$ is an overestimate or an underestimate.



Because $g''(-3) = -1$, we know the graph is concave down at $x = -3$ and can make a sketch of $g(x)$ like at the left. Then we can see that $g(-3.1)$ is on the tangent line and hence an overestimate.

Example 3:

- ① Show that $1 + kx$ is the local linearization of $(1 + x)^k$ at $x = 0$.
 - ② Someone claims that the square root of 1.1 is about 1.05. Without using a calculator, do you think that this estimate is about right?
 - ③ Is this approximation an overestimate or an underestimate? How can you tell from the graph?
 - ④ Find the error in this approximation.
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- ① Show that $1 + kx$ is the local linearization of $(1 + x)^k$ at $x = 0$.
Take the derivative: $\frac{d}{dx}(1 + x)^k = k(1 + x)^{k-1}$. Therefore, at $x = 0$, the slope is $k(1 + 0)^{k-1} = k$. Therefore the local linearization is the tangent line $y = k(x - 0) + 1 = kx + 1$ as desired.
- ② Someone claims that the square root of 1.1 is about 1.05. Without using a calculator, do you think that this estimate is about right?
Using the local linearization with $x = .1$ and $k = .5$, we have $\sqrt{1.1} = (1 + 0.1)^{.5} \approx .5(0.1) + 1 = 1.05$, so this estimate is reasonable.
- ③ Is this approximation an overestimate or an underestimate? How can you tell from the graph?
Since the graph of $\sqrt{x}$ is concave down, it is below the tangent line and the approximation from the tangent line is an overestimate.
- ④ Find the error in this approximation.
 $\sqrt{1.1} \approx 1.0488088\dots$, so the error (to 5 decimal places) is $1.05 - 1.048880 = 0.00119$.